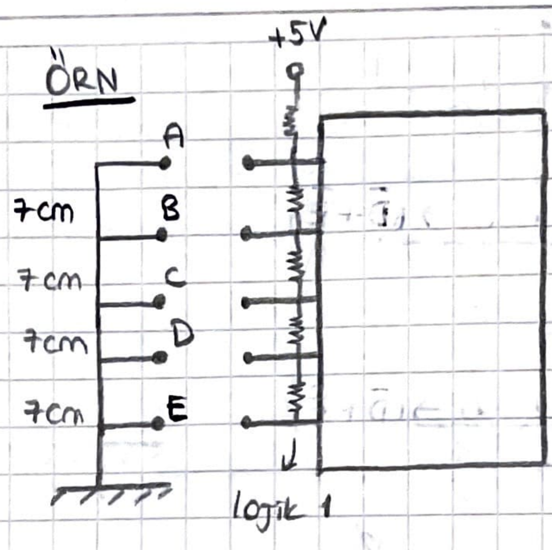
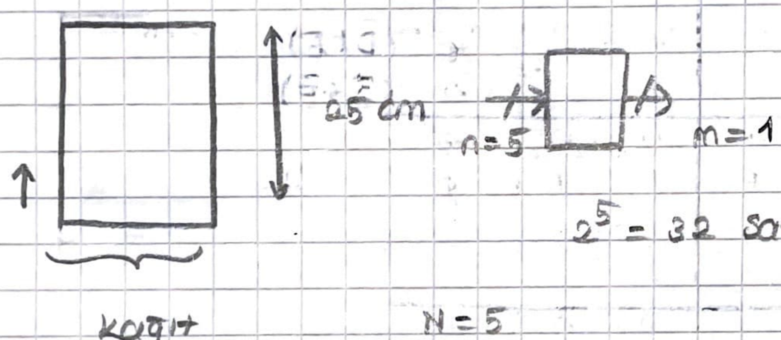
ÖRN




Şekilde bir fotokopi makinesi görülüyor. A, B, C, D ve E fotokopi makinesinde kağıdın yolu üzerinde sırasıyla dizilmiştir. A, B, C, D, E arası 7'şer cm. Anahtarlar normalde kapalıdır. Kağıt anahtarlar üzerinden geçerken açılıyor ve sonra tekrar kapanıyor. İki veya daha fazla anahtar açıkken lojik - 0 veren bir devreyi yalnızca NOR kapılarını kullanarak gerçekleştirin.

→ 3. mertebeden K-map kullanılacak




$2^5 = 32$ satır

N = 5
n = 3 ⇒ $N - n = 5 - 3 = 2$ → free variable

$2^2 = 4$ → 4'er 4'er böldük

Kağıt varsa → lojik 1
Kağıt yoksa → lojik 0

⇒ POS form yapılacak
lojik 0'lar indirgeniyor
} NOR kapısı

free variable

| | Dec | A | B | C | D | E | f | |
|---|---|---|---|---|---|---|---|---|
| | 0 | 0 | 0 | 0 | 0 | 0 | 1 | |
| | 1 | 0 | 0 | 0 | 0 | 1 | 1 | |
| 0 | 2 | 0 | 0 | 0 | 1 | 0 | Ø | |
| | 3 | 0 | 0 | 0 | 1 | 1 | 0 | → $(\bar{D}+\bar{E})$ |
| | 4 | 0 | 0 | 1 | 0 | 0 | Ø | |
| | 5 | 0 | 0 | 1 | 0 | 1 | Ø | |
| 1 | 6 | 0 | 0 | 1 | 1 | 0 | Ø | |
| | 7 | 0 | 0 | 1 | 1 | 1 | 0 | → $(\bar{D}+\bar{E})$ |
| | 8 | 0 | 1 | 0 | 0 | 0 | Ø | |
| | 9 | 0 | 1 | 0 | 0 | 1 | Ø | |
| 2 | 10 | 0 | 1 | 0 | 1 | 0 | Ø | Ø |
| | 11 | 0 | 1 | 0 | 1 | 1 | Ø | |
| | 12 | 0 | 1 | 1 | 0 | 0 | Ø | |
| | 13 | 0 | 1 | 1 | 0 | 1 | Ø | |
| 3 | 14 | 0 | 1 | 1 | 1 | 0 | 0 | $(\bar{D}+E)$ |
| | 15 | 0 | 1 | 1 | 1 | 1 | 0 | $(\bar{D}+\bar{E})$ |
| | 16 | 1 | 0 | 0 | 0 | 0 | 1 | |
| | 17 | 1 | 0 | 0 | 0 | 1 | Ø | |
| 4 | 18 | 1 | 0 | 0 | 1 | 0 | Ø | |
| | 19 | 1 | 0 | 0 | 1 | 1 | Ø | |
| | 20 | 1 | 0 | 1 | 0 | 0 | Ø | |
| | 21 | 1 | 0 | 1 | 0 | 1 | Ø | |
| 5 | 22 | 1 | 0 | 1 | 1 | 0 | Ø | Ø |
| | 23 | 1 | 0 | 1 | 1 | 1 | Ø | |
| | 24 | 1 | 1 | 0 | 0 | 0 | 0 | $(D+E)$ |
| | 25 | 1 | 1 | 0 | 0 | 1 | Ø | |
| 6 | 26 | 1 | 1 | 0 | 1 | 0 | Ø | |
| | 27 | 1 | 1 | 0 | 1 | 1 | Ø | |
| | 28 | 1 | 1 | 1 | 0 | 0 | 0 | $(D+E)$ |
| | 29 | 1 | 1 | 1 | 0 | 1 | Ø | |
| 7 | 30 | 1 | 1 | 1 | 1 | 0 | 0 | $(\bar{D}+E)$ |
| | 31 | 1 | 1 | 1 | 1 | 1 | Ø | |

kagıt varsa → 1
kağıt yoksa → 0

2 veya daha fazla anahter açık lojik 0

İndirgeme (1. yol) (Tavsiye edilmez, karışık)

| A \ BC | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | $\bar{D}+\bar{E}$ | $\bar{D}+\bar{E}$ | $(\bar{D}+\bar{E}).(\bar{D}+E)$ | $\emptyset$ |
| 1 | 1 | $\emptyset$ | $(D+E)(\bar{D}+E)$ | $D+$ |

İndirgeme (2. yol) → mantıklı, kolay

/0

| D \ E | 0 | 1 |
|---|---|---|
| 0 | 1 | 1 |
| 1 | $\emptyset$ | 0 |

$\bar{D}$

/1

| D \ E | 0 | 1 |
|---|---|---|
| 0 | $\emptyset$ | $\emptyset$ |
| 1 | $\emptyset$ | 0 |

$\bar{D}$

/2

| D \ E | | |
|---|---|---|
| 0 | $\emptyset$ | $\emptyset$ |
| 1 | $\emptyset$ | $\emptyset$ |

/3

| D \ E | 0 | 1 |
|---|---|---|
| 1 | $\emptyset$ | $\emptyset$ |
| 0 | 0 | 0 |

/4

| D \ E | 0 | 1 |
|---|---|---|
| 0 | 1 | $\emptyset$ |
| 1 | $\emptyset$ | $\emptyset$ |

$\bar{D}$

/5

| D \ E | 0 | 1 |
|---|---|---|
| 0 | $\emptyset$ | $\emptyset$ |
| 1 | $\emptyset$ | $\emptyset$ |

/6

| D \ E | 0 | 1 |
|---|---|---|
| 0 | 0 | $\emptyset$ |
| 1 | $\emptyset$ | $\emptyset$ |

/7

| D \ E | 0 | 1 |
|---|---|---|
| 0 | 0 | $\emptyset$ |
| 1 | 0 | $\emptyset$ |

→ $\emptyset$ ve 0'lı olanlara 0 koyduk. Çünkü don't care → 0/1

/f

| A \ BC | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | $\bar{D}$ | $\bar{D}$ | 0 | $\emptyset$ |
| 1 | $\bar{D}$ | $\emptyset$ | 0 | 0 |

$\bar{D}$

$\bar{B}$

$f = \bar{B}.\bar{D}$

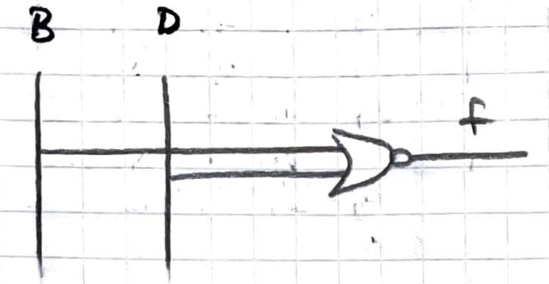
$0.\emptyset = 0$

$D.\bar{D}$

→ Hepsi birbirini götürüyor $\emptyset \to 0$, $0 \to D+\bar{D}$ alıyoruz.
Burdan da $\bar{D}$ seçersek, hepsi indirgenir $\bar{D}$ kalır.

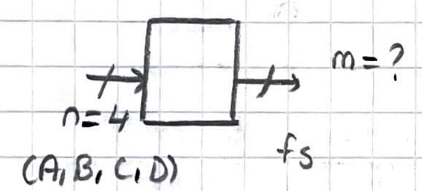
NOR kapılarıyla devrenin çizimi:

$$f = \bar{B}\,\bar{D} \rightarrow \bar{\bar{f}} = \overline{(B+D)}$$




**ÖRN** $f = x-5$ fonksiyonunu yalnızca NAND kapılarını kullanarak gerçekleştirin. X burada 4 bitlik bir değişkendir.

→ Fonksiyonun negatif değerleri kırmızı bir lamba ile gösterilecek.



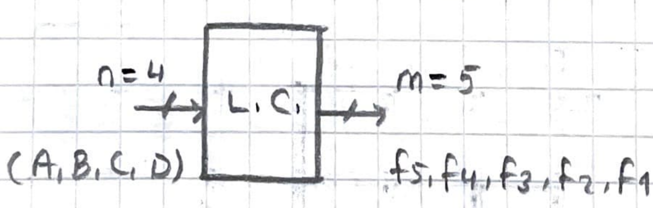

sign + magnitute
↓
$f_s$ : işaret biti

0 : pozitif demek
1 : negatif demek

Yani

| | s + m |
|---|---|
| +3 : | 0011 |
| -3 : | 1011 |

$$1011 \Rightarrow 0\cdot\cancel{2^3} + 0\cdot\cancel{2^2} + 1\cdot 2^1 + 1\cdot 2^0 = 3$$




$n=4$, $2^4 = 16$ satır

fonksiyonun degeri

(sign) işaret biti

magnitute (M)

| Dec | A | B | C | D | f=x-5 | $f_s$ | $f_3$ | $f_2$ | $f_1$ | $f_0$ |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | -5 | 1 | 0 | 1 | 0 | 1 |
| 1 | 0 | 0 | 0 | 1 | -4 | 1 | 0 | 1 | 0 | 0 |
| 2 | 0 | 0 | 1 | 0 | -3 | 1 | 0 | 0 | 1 | 1 |
| 3 | 0 | 0 | 1 | 1 | -2 | 1 | 0 | 0 | 1 | 0 |
| 4 | 0 | 1 | 0 | 0 | -1 | 1 | 0 | 0 | 0 | 1 |
| 5 | 0 | 1 | 0 | 1 | 0 | ∅ | 0 | 0 | 0 | 0 |
| 6 | 0 | 1 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 1 |
| 7 | 0 | 1 | 1 | 1 | 2 | 0 | 0 | 0 | 1 | 0 |
| 8 | 1 | 0 | 0 | 0 | 3 | 0 | 0 | 0 | 1 | 1 |
| 9 | 1 | 0 | 0 | 1 | 4 | 0 | 0 | 1 | 0 | 0 |
| 10 | 1 | 0 | 1 | 0 | 5 | 0 | 0 | 1 | 0 | 1 |
| 11 | 1 | 0 | 1 | 1 | 6 | 0 | 0 | 1 | 1 | 0 |
| 12 | 1 | 1 | 0 | 0 | 7 | 0 | 0 | 1 | 1 | 1 |
| 13 | 1 | 1 | 0 | 1 | 8 | 0 | 1 | 0 | 0 | 0 |
| 14 | 1 | 1 | 1 | 0 | 9 | 0 | 1 | 0 | 0 | 1 |
| 15 | 1 | 1 | 1 | 1 | 10 | 0 | 1 | 0 | 1 | 0 |

$f_s$

| AB \ CD | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 1 | 1 | 1 | 1 |
| 01 | 1 | ∅ | 0 | 0 |
| 11 | 0 | 0 | 0 | 0 |
| 10 | 0 | 0 | 0 | 0 |

$\bar{A}\bar{B}$

$\bar{A}\bar{C}$

$f_3$

| AB \ CD | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 0 | 0 | 0 | 0 |
| 01 | 0 | 0 | 0 | 0 |
| 11 | 0 | 1 | 1 | 1 |
| 10 | 0 | 0 | 0 | 0 |

$ABC$

$ABD$

$f_2$

| AB \ CD | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 1 | 1 | 0 | 0 |
| 01 | 0 | 0 | 0 | 0 |
| 11 | 1 | 0 | 0 | 0 |
| 10 | 0 | 1 | 1 | 1 |

$\bar{A}\bar{B}\bar{C}$

$AB\bar{C}\bar{D}$

$A\bar{B}D$

$A\bar{B}C$

$f_1$

| AB \ CD | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 0 | 0 | 1 | 1 |
| 01 | 0 | 0 | 1 | 0 |
| 11 | 1 | 0 | 1 | 0 |
| 10 | 1 | 0 | 1 | 0 |

$\bar{A}\bar{B}C$

$A\bar{C}\bar{D}$

$CD$

| AB \ CD | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 1 | 0 | 0 | 1 |
| 01 | 1 | 0 | 0 | 1 |
| 11 | 1 | 0 | 0 | 1 |
| 10 | 1 | 0 | 0 | 1 |

$\to \bar{D}$

$f_0$

Yani;

$f_5 = \bar{A}\bar{B} + \bar{A}\bar{C}$

$f_3 = ABD + ABC$

$f_2 = \bar{A}\bar{B}\bar{C} + A\bar{B}D + A\bar{B}C + AB\bar{C}\bar{D}$

$f_1 = A\bar{C}\bar{D} + \bar{A}\bar{B}C + CD$

$f_0 = \bar{D}$

NAND dediği için $=$ alıyoruz;

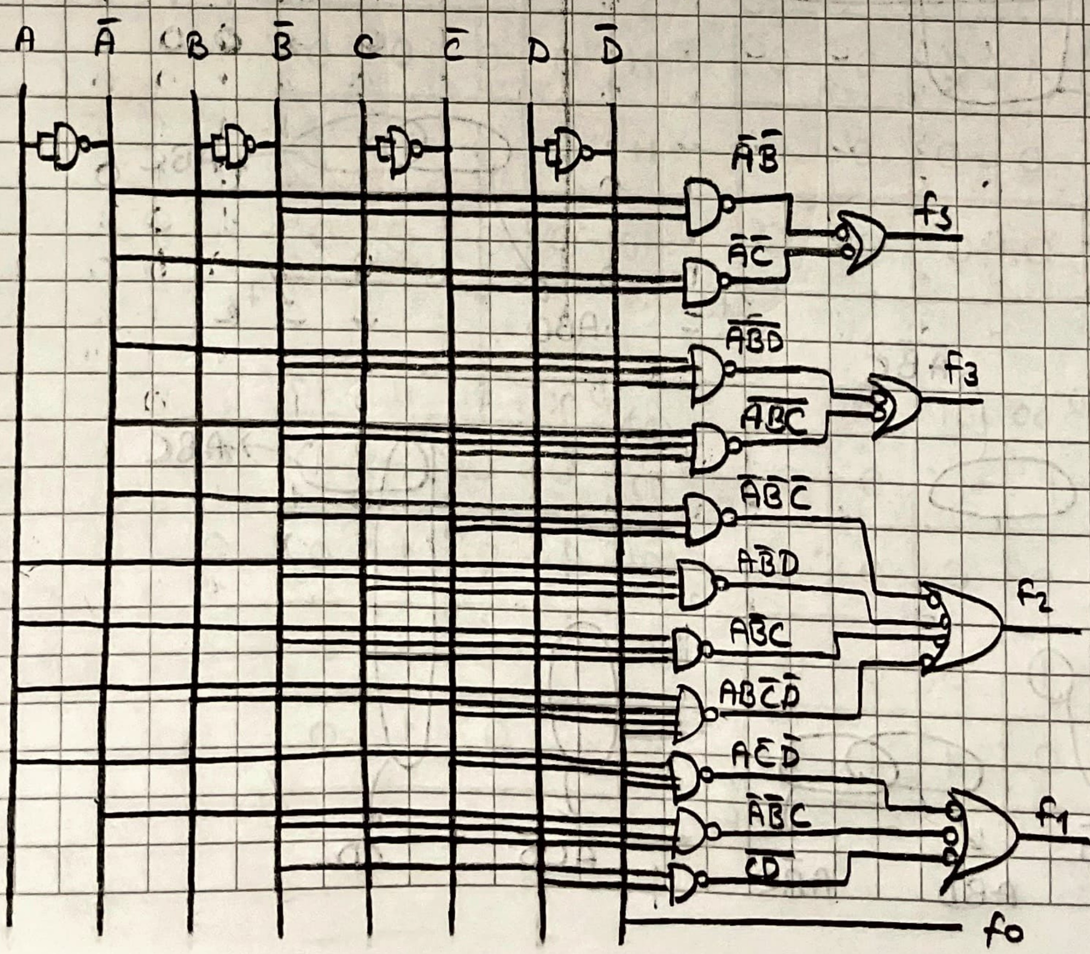
$\bar{\bar{f_5}} = \overline{\overline{(\bar{A}\bar{B}) + (\bar{A}\bar{C})}} = \overline{\overline{(\bar{A}\bar{B})} \cdot \overline{(\bar{A}\bar{C})}}$

$\bar{\bar{f_3}} = \overline{\overline{(ABD) + (ABC)}} = \overline{\overline{(ABD)} \cdot \overline{(ABC)}}$

$\bar{\bar{f_2}} = \overline{\overline{\bar{A}\bar{B}\bar{C} + A\bar{B}D + A\bar{B}C + AB\bar{C}\bar{D}}} = \overline{\overline{(\bar{A}\bar{B}\bar{C})} \cdot \overline{(A\bar{B}D)} \cdot \overline{(A\bar{B}C)} \cdot \overline{(AB\bar{C}\bar{D})}}$

$\bar{\bar{f_1}} = \overline{\overline{A\bar{C}\bar{D} + \bar{A}\bar{B}C + CD}} = \overline{\overline{(A\bar{C}\bar{D})} \cdot \overline{(\bar{A}\bar{B}C)} \cdot \overline{(CD)}}$

$f_0 = \bar{D}$

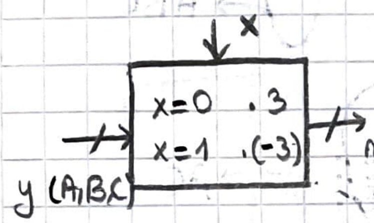


**ÖRN** Bir lojik devrede y(A,B,C) 3 bitlik binary veri girişi, x ise kontrol girişidir.

x = 0 olduğunda y, +3 ile
x = 1 olduğunda y, -3 ile çarpılmalıdır.

Böyle bir devreyi yalnızca NAND kapılarını kullanarak gerçekleştirin. (Minimum maliyetle)

→ Fonksiyonun işareti için negatif değerler → lojik 1
pozitif değerler → lojik 0




m=6.

7.3 → 21
7.(-3) → 21    5 bit ($2^4$)
(4 3 2 1 0)

| | Dec x | Dec y | x | y: A | y: B | y: C | f | $f_5$ | $f_4$ (16) | $f_3$ (8) | $f_2$ (4) | $f_1$ (2) | $f_0$ (1) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | ∅ | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 | 0 | 0 | 1 | +3 | 0 | 0 | 0 | 0 | 1 | 1 |
| 2 | 0 | 2 | 0 | 0 | 1 | 0 | +6 | 0 | 0 | 0 | 1 | 1 | 0 |
| 3 | 0 | 3 | 0 | 0 | 1 | 1 | +9 | 0 | 0 | 1 | 0 | 0 | 1 |
| 4 | 0 | 4 | 0 | 1 | 0 | 0 | +12 | 0 | 0 | 1 | 1 | 0 | 0 |
| 5 | 0 | 5 | 0 | 1 | 0 | 1 | +15 | 0 | 0 | 1 | 1 | 1 | 1 |
| 6 | 0 | 6 | 0 | 1 | 1 | 0 | +18 | 0 | 1 | 0 | 0 | 1 | 0 |
| 7 | 0 | 7 | 0 | 1 | 1 | 1 | +21 | 0 | 1 | 0 | 1 | 0 | 1 |
| 8 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | ∅ | 0 | 0 | 0 | 0 | 0 |
| 9 | 1 | 1 | 1 | 0 | 0 | 1 | -3 | 1 | 0 | 0 | 0 | 1 | 1 |
| 10 | 1 | 2 | 1 | 0 | 1 | 0 | -6 | 1 | 0 | 0 | 1 | 1 | 0 |
| 11 | 1 | 3 | 1 | 0 | 1 | 1 | -9 | 1 | 0 | 1 | 0 | 0 | 1 |
| 12 | 1 | 4 | 1 | 1 | 0 | 0 | -12 | 1 | 0 | 1 | 1 | 0 | 0 |
| 13 | 1 | 5 | 1 | 1 | 0 | 1 | -15 | 1 | 0 | 1 | 1 | 1 | 1 |
| 14 | 1 | 6 | 1 | 1 | 1 | 0 | -18 | 1 | 1 | 0 | 0 | 1 | 0 |
| 15 | 1 | 7 | 1 | 1 | 1 | 1 | -21 | 1 | 1 | 0 | 1 | 0 | 1 |

| XA \ BC | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | ∅ | 0 | 0 | 0 |
| 01 | 0 | 0 | 0 | 0 |
| 11 | 1 | 1 | 1 | 1 |
| 10 | ∅ | 1 | 1 | 1 |

$\to X$

$f_5$

| XA \ BC | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 0 | 0 | 0 | 0 |
| 01 | 0 | 0 | 1 | 1 |
| 11 | 0 | 0 | 1 | 1 |
| 10 | 0 | 0 | 0 | 0 |

$\to AB$

$f_4$

| XA \ BC | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 0 | 0 | 1 | 0 |
| 01 | 1 | 1 | 0 | 0 |
| 11 | 1 | 1 | 0 | 0 |
| 10 | 0 | 0 | 1 | 0 |

$\to \bar{A}BC$

$A\bar{B}$

$f_3$

| XA \ BC | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 0 | 0 | 0 | 1 |
| 01 | 1 | 1 | 1 | 0 |
| 11 | 1 | 1 | 1 | 0 |
| 10 | 0 | 0 | 0 | 1 |

$\to \bar{A}B\bar{C}$

$\to AC$

$A\bar{B}$

$f_2$

| XA \ BC | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 0 | 1 | 0 | 1 |
| 01 | 0 | 1 | 0 | 1 |
| 11 | 0 | 1 | 0 | 1 |
| 10 | 0 | 1 | 0 | 1 |

$f_1$

$\bar{B}C$ $B\bar{C}$

| AX \ BC | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 0 | 1 | 1 | 0 |
| 01 | 0 | 1 | 1 | 0 |
| 11 | 0 | 1 | 1 | 0 |
| 10 | 0 | 1 | 1 | 0 |

$\to C$

$f_0$

$f_5 = X$

$f_4 = AB$

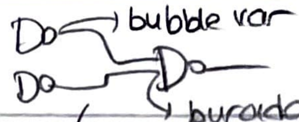
$f_3 = A\bar{B} + \bar{A}BC$

$f_2 = A\bar{B} + AC + \bar{A}B\bar{C}$

$f_1 = \bar{B}C + B\bar{C}$

$f_0 = C$

Logic uyumsuzluk →




Bu bağlantının ya iki tarafında bubble olacak ya da iki tarafında da olmayacak.

→ NAND kapılarıyla gerçekleştirme


$f_5 = X$

$\overline{\overline{f_4}} = \overline{\overline{AB}}$

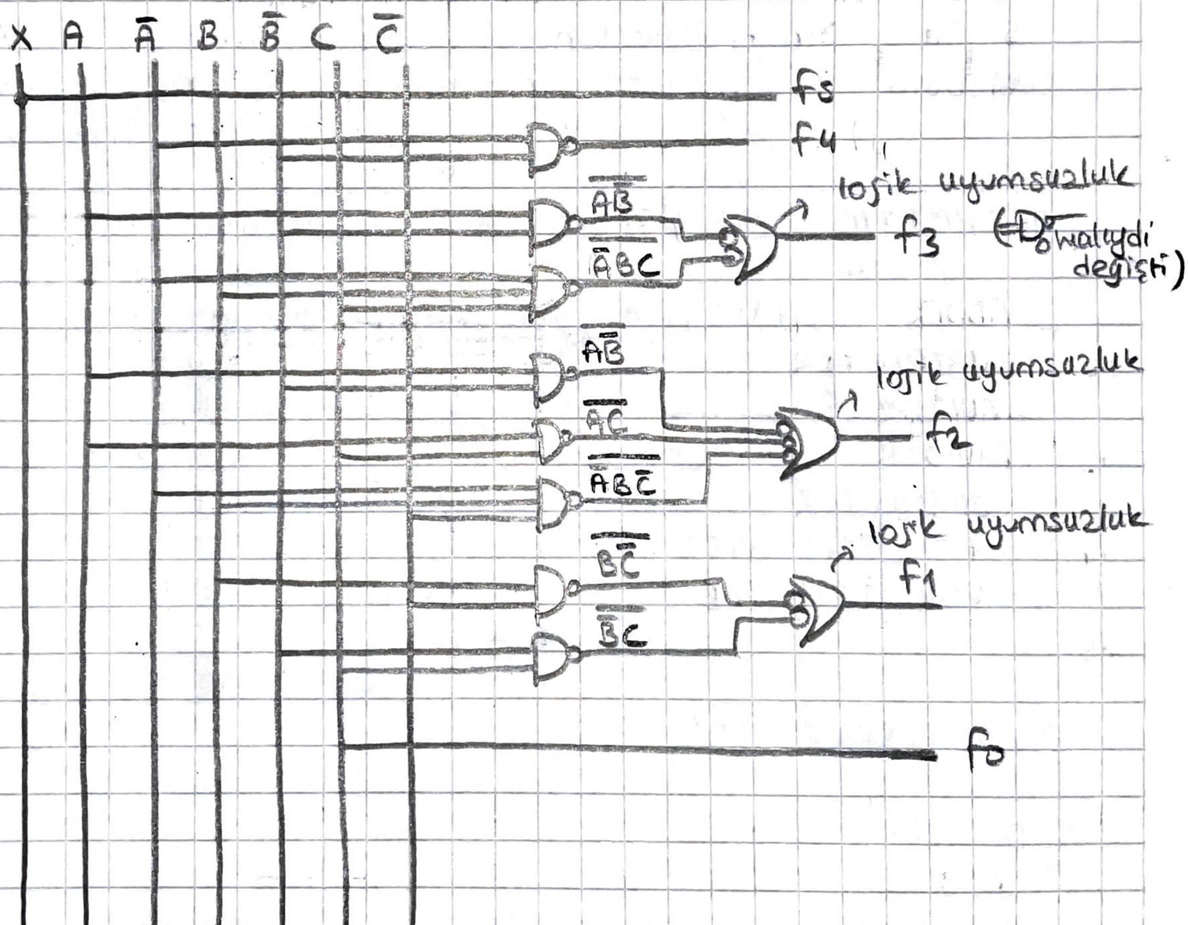
$\overline{\overline{f_3}} = \overline{\overline{A\overline{B} + \overline{A}BC}} = \overline{(\overline{A\overline{B}}) \cdot (\overline{\overline{A}BC})}$

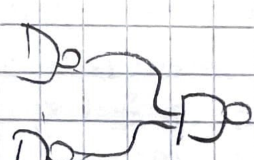
$\overline{\overline{f_2}} = \overline{\overline{A\overline{B} + AC + \overline{A}B\overline{C}}} = \overline{(\overline{A\overline{B}}) \cdot (\overline{AC}) \cdot (\overline{\overline{A}B\overline{C}})}$

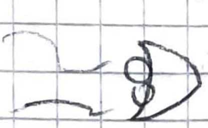
$f_1 = \overline{\overline{B\overline{C} + \overline{B}C}} = \overline{(\overline{B\overline{C}}) \cdot (\overline{\overline{B}C})}$

$(B \oplus C)$ ?

$f_0 = C$

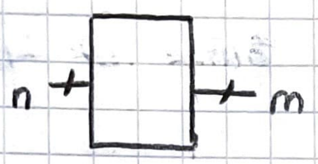


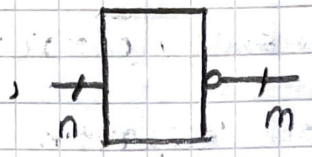





COMBINATIONAL LOGIC DESIGN

1) Gate Perspective (Approach)
2) Modular Perspective (Approach)




THE BUILDING BLOCKS

1) Data Manipulation
2) Data Code Conversion
3) Data Selection from Various Sources
4) Data Routing from Source to Various Destinctions
5) Data Error Detection
6) Data Busing from one Part of The Digital System to Another

ARITHMETIC TYPE CIRCUIT

ADDER
SUBTRACTOR
MULTIPLIER
DIVIDER
COMPARATOR
PARITY GENERATOR
ERROR DETECTOR

MULTI PURPOSE CIRCUIT

DECODER - ENCODER
CODE CONVERTER
MULTIPLEXER DEMULTIPLEXER
SHIFTER
ALU (Arithmetic and Logic Unit)
ROM (Read only memory)
PLA (Programmable Logic Array)
PAL (Programmable Array Logic)

# CLASSIFICATION OF IC CHIPS

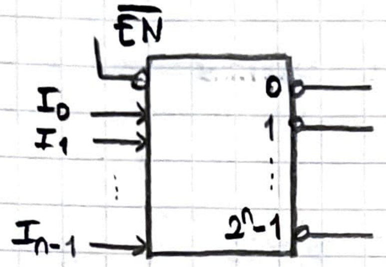
## IC Destiny of Integration

| Destiny of Integration / Complexity | Gates per Ic |
|---|---|
| SSI : Small Scale Integration<br>• Logic Gates (AND, OR, NAND,...) | < 10 |
| MSI : medium Scale Integration<br>• Flip Flop<br>• Adders / Counters<br>• Multiplexers / Demultiplexers | 10 - 100 |
| LSI : Large Scale Integration<br>• Small memory Chips<br>• Programmable logic Device | 100 - 10.000 |
| VLSI : Very large scale Integration<br>• large memory chips<br>• complex Programmable Logic Device | 10.000 - 100.000 |
| ULSI : Ultra Large Scale Integration<br>• 8 & 16 Bit microprocessors | 100.000 - 1.000.000 |
| GSI : Giga Scale Integration<br>• Pentium IV Processors | |

## DESIGN PROCEDURE

1. Understand the Device
2. State the Algorithm
3. Construct a truth Table
4. Obtain Output Function
5. Construct the Logic Diagram
6. Check (Test) the Results

# DECODER AND ENCODER




$y_0(L) = m_0 \, EN(L)$

$y_1(L) = m_1 \, EN(L)$

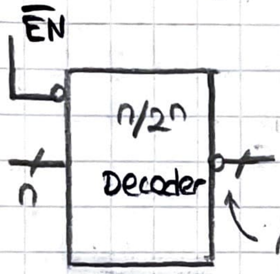
$\vdots$

$y_{2^n-1}(L) = m_{2^n-1} \, EN(L)$

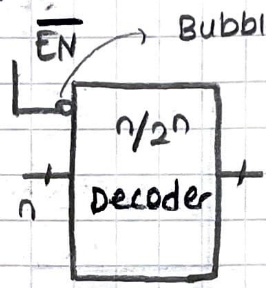
$m_i(L) = M_i(H)$

$M_i(L) = m_i(H)$

$y_i = m_i \, EN(L)$




Active low Output (0'da Aktif)

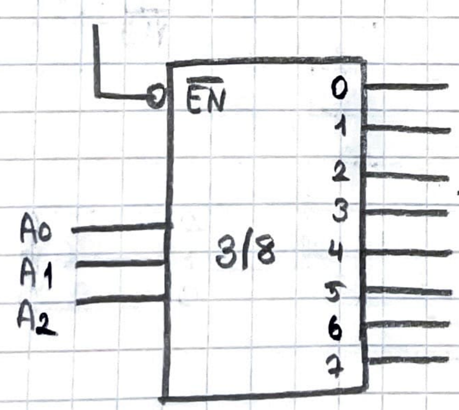



Bubble ucu 0'da aktif çalışır

Active High Output

| | $\overline{EN}$ | $A_2$ | $A_1$ | $A_0$ | $y_7$ | $y_6$ | $y_5$ | $y_4$ | $y_3$ | $y_2$ | $y_1$ | $y_0$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | x | x | x | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| $m_0$ | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| $m_1$ | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 |
| $m_2$ | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |
| $m_3$ | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| $m_4$ | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 |
| $m_5$ | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| $m_6$ | 0 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| $m_7$ | 0 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

$y_i = m_i \, EN(L)$




$m_0 \, EN(L) \} \; \overline{A_2}\,\overline{A_1}\,\overline{A_0} \, EN(L)$

minterm

EN(0)

EN(1) → Aktifleştirme

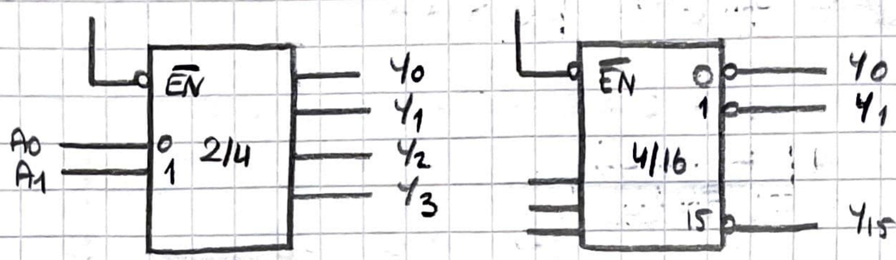
$m_7 \, EN(L) \} \; A_2 A_1 A_0 \, EN(L)$

→ n bitlik bir sözcüğün kodunu çözüp olası $2^n$ çıkıştan sadece birini aktif hale getiren kombinezonal devredir.

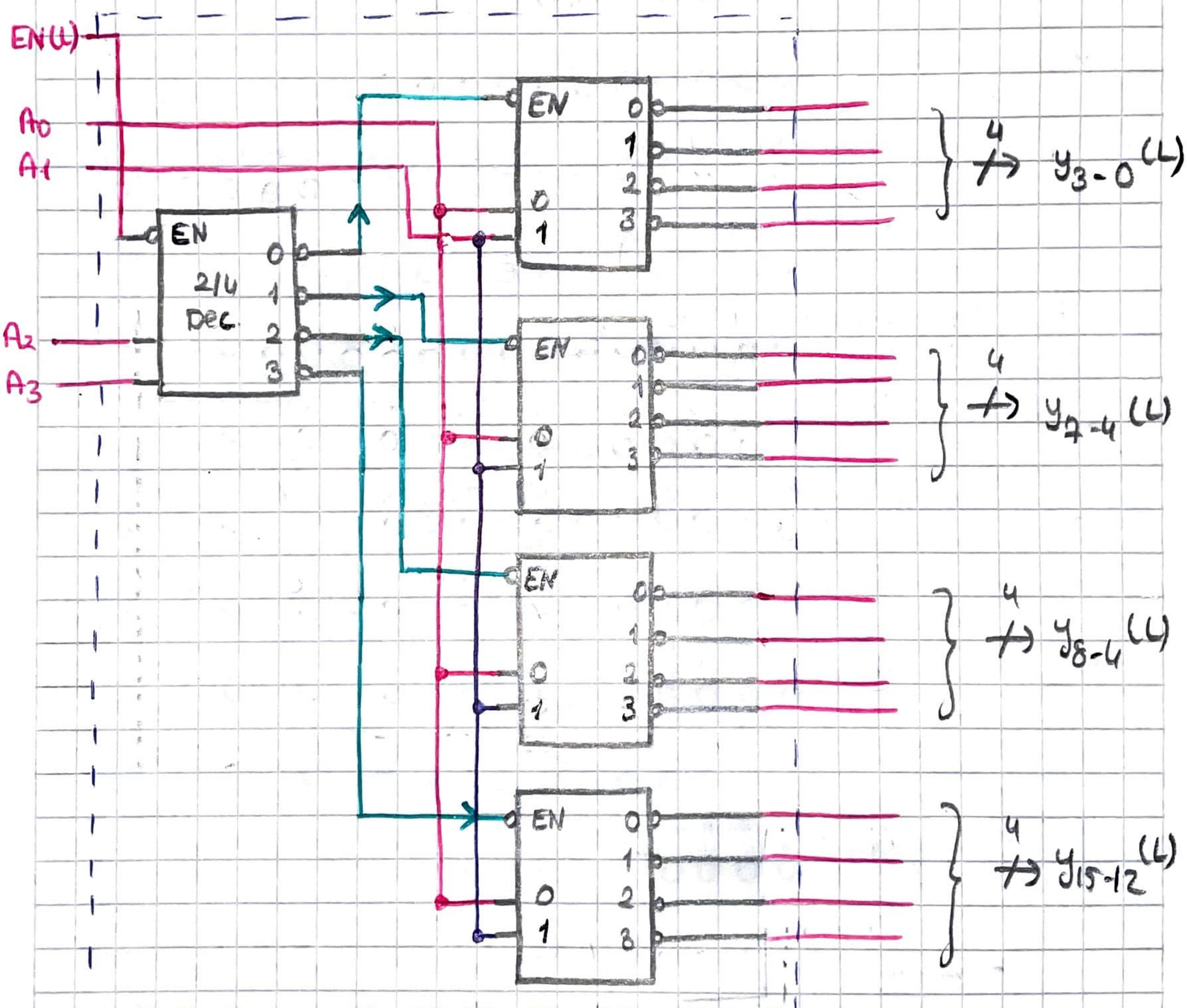
100 daireli apart. zil sistemi
mux
demux ✓ decoder
✓ encoder | yapılabilir

→ $n=3$ için decoderin $2^3 = 8$ çıkışı bulunur.

EXPANSION FORMAT




2 → 4 4/16 Decoder

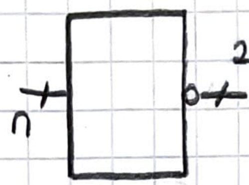



| EN | $A_1$ | $A_0$ | $y_3$ | $y_2$ | $y_1$ | $y_0$ |
|---|---|---|---|---|---|---|
| 1 | x | x | 1 | 1 | 1 | 1 |
| 0 | 0 | 0 | 1 | 1 | 1 | 0 |
| 0 | 0 | 1 | 1 | 1 | 0 | 1 |
| 0 | 1 | 0 | 1 | 0 | 1 | 1 |
| 0 | 1 | 1 | 0 | 1 | 1 | 1 |

| $A_3$ | $A_2$ | $A_1$ | $A_0$ | $y_{15}$ | $y_{14}$ | $y_{13}$ | $y_{12}$ | $y_{11}$ | $y_{10}$ | $y_9$ | $y_8$ | $y_7$ | $y_6$ | $y_5$ | $y_4$ | $y_3$ | $y_2$ | $y_1$ | $y_0$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 |
| 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |
| 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 |
| 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

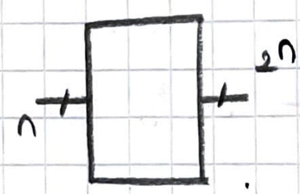
# COMBINATIONAL LOGIC DESIGN WITH DECODER

## DECODER




inverting output

→ Inverting output varsa bunun çıkışı maxterm üretir.




noninverting output

→ noninverting output varsa çıkışı minterm üretir.

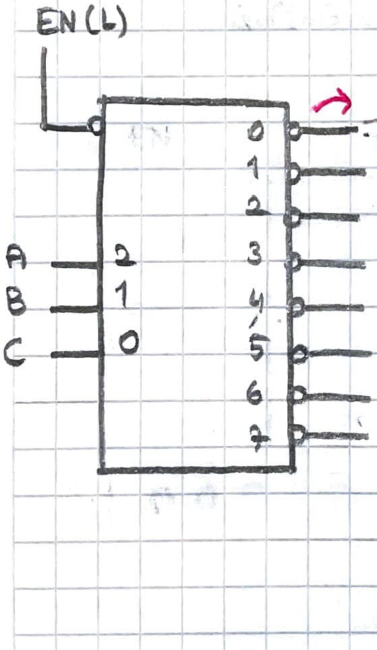
→ Decoder kullanabilmek için fonksiyonun cannonical formda olması gerekir.

Logic Function

→ cannonical SOP Form
→ cannonical POS Form

**ÖRN** $F(A,B,C) = \Sigma m(1,3,4,7)$
$G(A,B,C) = \Pi M(0,2,5,6)$

Eviren çıkışlı decoder (3/8 Decoder) kullanarak bu fonkları gerçekleştiriniz.




→ Çıkışı inverting olduğu için maxterm üretir.

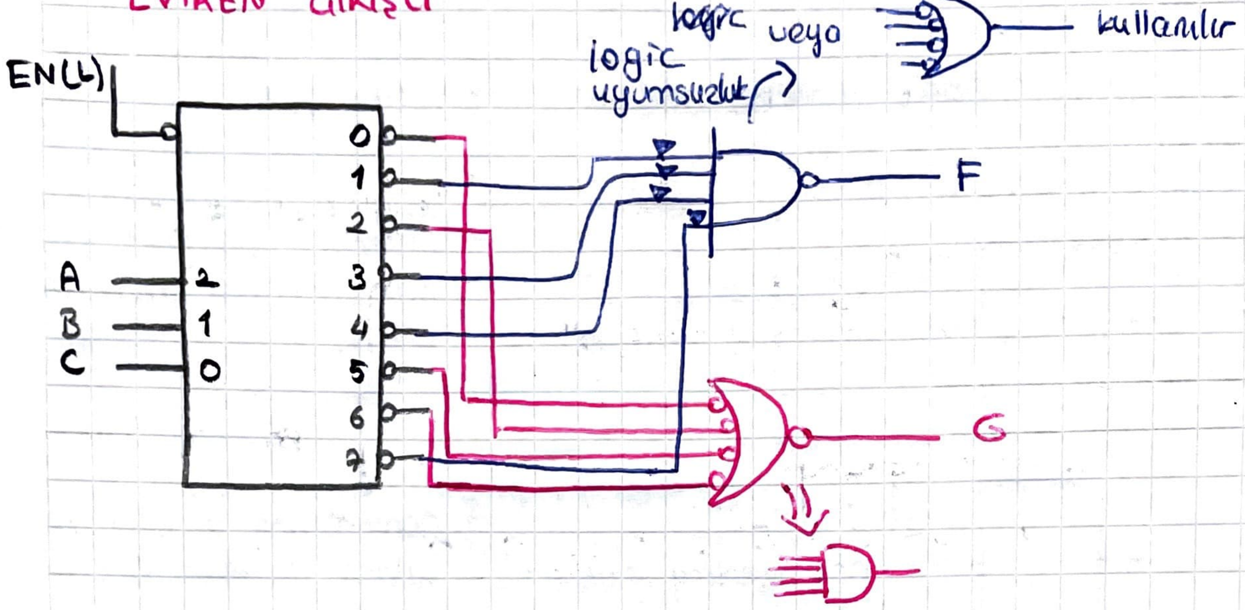
$\overline{m_i} = M_i$

$\overline{\overline{F}} = \overline{\overline{m_1 + m_3 + m_4 + m_7}} = \overline{m_1} \cdot \overline{m_3} \cdot \overline{m_4} \cdot \overline{m_7}$

$\overline{\overline{F}} = \overline{M_1 \cdot M_3 \cdot M_4 \cdot M_7}$

$G = M_0 \cdot M_2 \cdot M_5 \cdot M_6$

## EVİREN ÇIKIŞLI




∇ Eviren → ~~POS~~ SOP

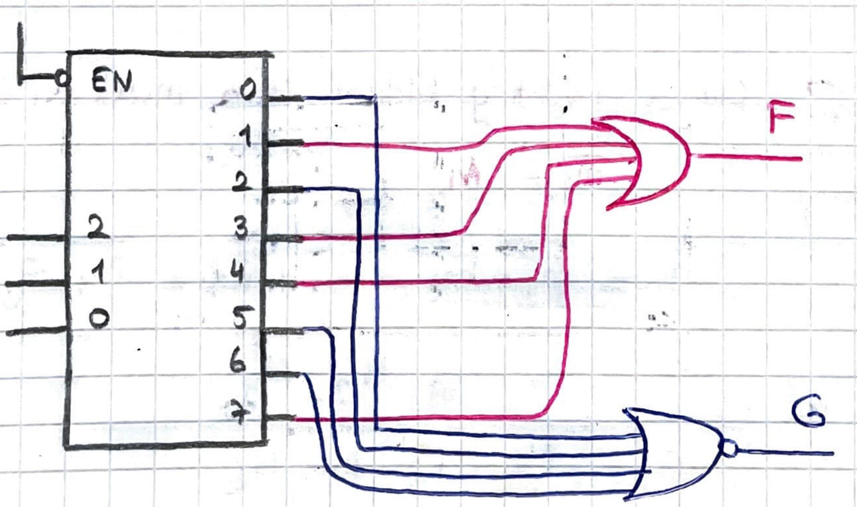
Evirmeyen → ~~SOP~~ POS

## EVİRMEYEN ÇIKIŞLI

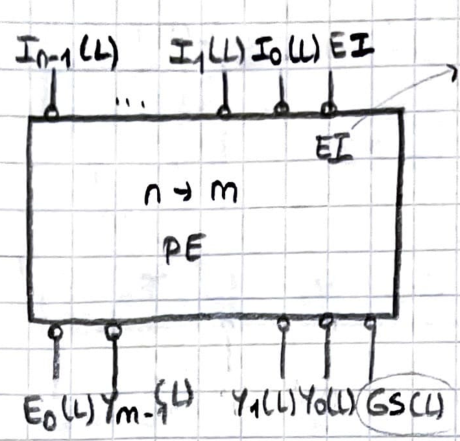
$F = m_1 + m_3 + m_4 + m_7$

$G = \overline{M_0 \cdot M_2 \cdot M_5 \cdot M_6} = \overline{\overline{M_0} + \overline{M_2} + \overline{M_5} + \overline{M_7}}$

$= \overline{m_0 + m_2 + m_5 + m_7}$



PRIORITY ENCODER (PE)




→ Enable Input (Etkinleştirme Girişi)

$n \leq 2^m$ } Design of Priority Encoder

GS(L) → Group select

| Priority schedule | $\overline{EI}$ | $I_2$ | $I_1$ | $I_0$ | GS | $Y_1$ | $Y_0$ | $E_0$ |
|---|---|---|---|---|---|---|---|---|
| | 0 | x | x | x | 0 | 0 | 0 | 0 |
| $I_2$ Encoded to 11 | 1 | 1 | x | x | 1 | 1 | 1 | 0 |
| $I_1$ Encoded to 10 | 1 | 0 | 1 | x | 1 | 1 | 0 | 0 |
| $I_0$ Encoded to 01 | 1 | 0 | 0 | 1 | 1 | 0 | 1 | 0 |
| Inactive state to 00 Designed | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |

$Y_0$

| $I_2$ \ $I_1 I_0$ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | 0 | EI | 0 | 0 |
| 1 | EI | EI | EI | EI |

$Y_1$

| $I_2$ \ $I_1 I_0$ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | 0 | 0 | EI | EI |
| 1 | EI | EI | EI | EI |

$E_0$

| $I_2$ \ $I_1 I_0$ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | EI | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 0 |

GS

| $I_2$ \ $I_1 I_0$ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | 0 | EI | EI | EI |
| 1 | EI | EI | EI | EI |

$$y_0 = \overline{I_1}\, I_0\, EI + I_2\, EI$$

$$y_1 = I_2\, EI + I_1\, EI$$

$$E_0 = \overline{I_2}\,\overline{I_1}\,\overline{I_0}\, EI$$

$$GS = (I_2 + I_1 + I_0)\, EI = \overline{E_0}\, EI$$

| $EI$ | $I_5$ | $I_4$ | $I_3$ | $I_2$ | $I_1$ | $I_0$ | $GS$ | $Y_1$ | $Y_0$ | $E_0$ |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | x | x | x | x | x | x | 0 | 0 | 0 | 0 |
| 1 | 1 | x | x | x | x | x | 1 | 1 | 1 | 0 |
| 1 | 0 | 1 | x | x | x | x | 1 | 1 | 0 | 0 |
| 1 | 0 | 0 | 1 | x | x | x | 1 | 0 | 1 | 0 |
| 1 | 0 | 0 | 0 | 1 | x | x | 0 | 1 | 1 | 0 |
| 1 | 0 | 0 | 0 | 0 | 1 | x | 0 | 1 | 0 | 0 |
| 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |